ISEN/DAEN 427 Introduction (Still)

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Bayesian Modeling

Deduction	**Induction**
Theory	Observation
↓	↓
Hypothesis	Pattern
↓	↓
Observation	Hypothesis
↓	↓
Confirmation	Theory

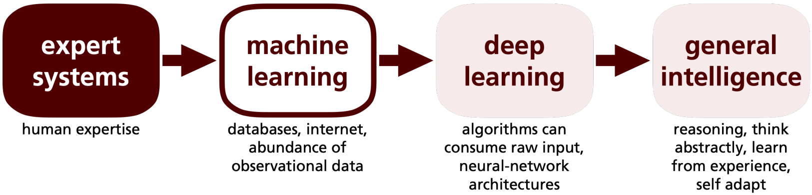
Aristotle

Sherlock

Model Building

- Spoken/written language
- Conceptual diagrams
- Mathematical notation
- Computer code

Data and Models (forms and functions)



	expert systems	machine learning	deep learning	general intelligence
DATA	human	+ databases	+ sensory	+ everything
EXEMPLAR	rules	rules/networks	neural networks	pre-trained deep neural networks
SCOPE	follows	+ discover relationships	+ senses relationships	+ understands the world
CURATION	high	medium	low	minimal

Interpreting probabilities:

- **Frequentist interpretation**:
  Defines probability as the long-run frequency of an event in repeated, identical trials. It treats parameters as fixed and only data as random. Probabilities are objective and apply only to repeatable events.
- **Subjective or Bayesian interpretation**:
  Views probability as a degree of belief, updated with new evidence using Bayes' theorem. It allows assigning probabilities to hypotheses and parameters. Probabilities are subjective and reflect uncertainty.

An Essay Towards Solving a Problem in the Doctrine of Chances by Thomas Bayes (1763)

- Aimed to determine the probability (chances) of a cause given an observed event.
- First formal attempt to update beliefs using new evidence.
- Challenged the classical view of probability ("doctrine of chances") by introducing a method to handle uncertainty about unknown parameters using prior beliefs and observed data.
- Although not recognized during Bayes' lifetime [1], the essay laid the groundwork for modern statistical reasoning, especially in fields where data is incomplete or uncertain.

[1] Published two years after Bayes' death, containing multiple amendments and additions due to his friend Richard Price.

The Logic of Chance by John Venn (1866)

What is subjective probability?

- Based on personal belief or degree of confidence in an event's occurrence (not derived from frequency or objective data).
- Can vary between individuals based on experience, information, or intuition.

Venn's view

- Critical of purely subjective views (while he acknowledged human judgment, he believed probability must rest on empirical evidence, not personal belief alone).
- Acknowledged subjective reasoning but saw it as insufficient for scientific analysis.

Bayesian estimation of cow weight

Bayesian estimation of cow weight

- Prior belief: the cow weighs 500 ± 100 kg

Bayesian estimation of cow weight

- Prior belief: the cow weighs 500 ± 100 kg
- Survey guesses: $400, 450, 600, 900, 1000$

Step 1: Prior Distribution

Assume the prior distribution is normal:

$$\theta \sim \mathcal{N}(\mu_0 = 500, \sigma_0^2 = 100^2)$$

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Step 2: Likelihood Model

Assume each guess x_i is an independent observation with Gaussian noise:

$$x_i \sim \mathcal{N}(\theta, \sigma^2), \quad \text{where } \sigma = 100$$

Survey data:

$$x = [400, 450, 600, 900, 1000]$$

$$\bar{x} = \frac{400 + 450 + 600 + 900 + 1000}{5} = \frac{3350}{5} = 670$$

Step 3: Posterior Distribution

We use conjugate prior updating (Normal-Normal model):

$$\mu_n = \frac{\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}}, \quad \sigma_n^2 = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}\right)^{-1}$$

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Substitute known values:

$$\mu_n = \frac{\frac{500}{100^2} + \frac{5 \cdot 670}{100^2}}{\frac{1}{100^2} + \frac{5}{100^2}} = \frac{500 + 3350}{6} = \frac{3850}{6} \approx 641.67$$

$$\sigma_n^2 = \left(\frac{1}{100^2} + \frac{5}{100^2} \right)^{-1} = \left(\frac{6}{100^2} \right)^{-1} = \frac{100^2}{6} \approx 1666.67$$

$$\sigma_n \approx \sqrt{1666.67} \approx 40.8$$

The posterior distribution is:

$$\boxed{\theta \mid x \sim \mathcal{N}(642, 41^2)}$$

After seeing the data,
best estimate shifts from 500 kg $\rightarrow$ 642 kg,
and uncertainty reduces from $\pm 100 \rightarrow \pm 41$ kg.

Bayesian Decision Analysis

- Using probability to make better decisions under uncertainty: *diachronic interpretation*.
- Interpretation that takes into account the historical change or temporal development of a phenomenon, idea, language, concept, etc.

  H: Hypothesis

  D: Data

  Given $p(H)$, prob of the hypothesis before the data;

  find $p(H \mid D)$, prob of the hypothesis after the data.

- Bayes's theorem is a recipe for updating beliefs in light of new data. Then we can use beliefs to guide decisions.
- If you are estimating a quantity, and a point estimate is good enough, use whatever.
- But if you are making decisions, Bayesian methods guide decision-making under uncertainty; frequentist methods don't.

$$p(H \mid D) = p(H)\, p(D \mid H) \,/\, p(D)$$

posterior

prior

likelihood

total probability
of the data

$$p(H \mid D) \sim p(H)\, p(D \mid H)$$

posterior

prior

likelihood

normalize
the posterior probs

$\sim$ ("tilde") here means "approximately equal to"

Cromwell's rule:

- *"I beseech you, in the bowels of Christ, think it possible that you may be mistaken."*
- Alters the first rule of probability (the convexity rule) from $0 \leq Pr(\cdot) \leq 1$ to $0 < Pr(\cdot) < 1$.

Bayesian Models

Probability theory is nothing but common sense reduced to calculation. *Pierre Simon Laplace*

- Unknown quantities are described using **probability distributions**. We call these quantities **parameters**.
- Bayes' theorem is used to update the values of the parameters **conditioned on the data**. We can also see this process as a reallocation of probabilities.

Bayesian models as **generative models**:

... are like storytelling tools that explain how data could come from some unknown hidden things with uncertainty included.

- Learn the joint probability distribution of inputs and outputs, allowing them to generate new, realistic data samples.
- First, you guess some hidden factors or settings (called parameters) before seeing any data: this is the **prior** belief.
- Then, based on those hidden factors, you explain how the actual data could be produced: this is the **likelihood**.

Bayesian Workflow (steps in an iterative non-linear fashion)

1. Given some data and some assumptions on how this data could have been generated, design a model by combining and transforming random variables.
2. Use Bayes to condition models to the available data.
   - This process is called **inference**, and as a result it obtains a posterior distribution. Hope the data reduces the uncertainty for possible parameter values (though this is not a guarantee of any Bayesian model).
3. Criticize the model by checking whether the model makes sense according to different criteria, including the data and expertise on the domain-knowledge. Because generally there are uncertain about the models themselves, compare several models.

Bayesian Inference

- In colloquial terms, inference is associated with obtaining **conclusions** based on *evidence* and *reasoning*.
- Bayesian inference is a particular form of statistical inference based on combining probability distributions in order to obtain **other probability distributions**.

Bayes' theorem:

$$\underbrace{p(\boldsymbol{\theta} \mid \boldsymbol{Y})}_{\text{posterior}} = \frac{\overbrace{p(\boldsymbol{Y} \mid \boldsymbol{\theta})}^{\text{likelihood}} \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\boldsymbol{Y})}_{\text{marginal likelihood}}}$$

$$p(\boldsymbol{Y}) = \int_{\boldsymbol{\Theta}} p(\boldsymbol{Y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

$$\underbrace{p(\boldsymbol{\theta} \mid \boldsymbol{Y})}_{\text{posterior}} \propto \overbrace{p(\boldsymbol{Y} \mid \boldsymbol{\theta})}^{\text{likelihood}} \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}$$

$\propto$ it is not a "half infinity" but means "is proportional to"

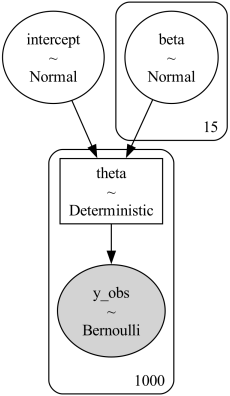
Example:

Let us try to solve the Beta-Binomial model.

This is probably the most common example in Bayesian statistics and it is used to model binary, mutually-exclusive outcomes such as 0 or 1, positive or negative, head or tails, spam or ham, healthy or unhealthy, etc.

Statistical notation for the Beta-Binomial model is given by:

$$\theta \sim \text{Beta}(\alpha, \beta) \text{ represents the prior distribution}$$
$$Y \sim \text{Bin}(n = 1, p = \theta) \text{ represents the likelihood distribution}$$

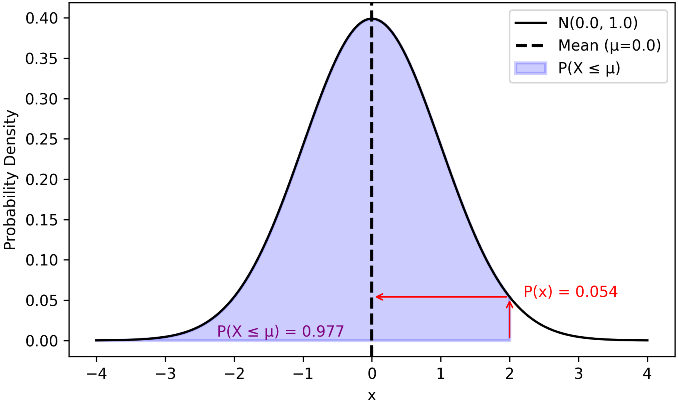

intercept
~
Normal
beta
~
Normal
15
theta
~
Deterministic
y_obs
~
Bernoulli
1000

The probability density function (PDF) of the standard normal distribution is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

Evaluating the PDF at $z = 2$: $f(2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{2^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-2} \approx 0.05399$

So the likelihood at $z = 2$ is approximately ≈ 0.054

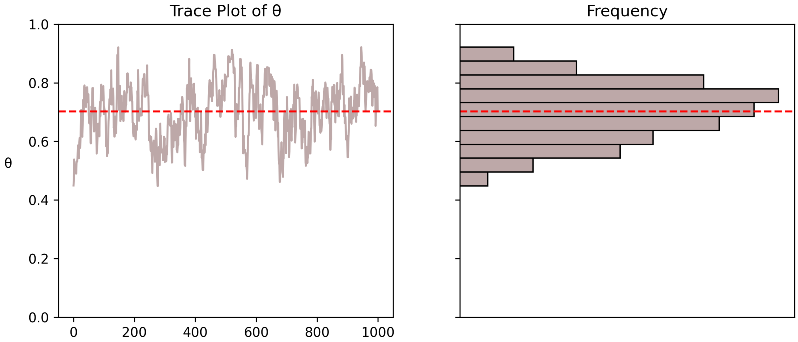


Alternatively, if you meant the cumulative probability: $P(Z \leq 2) \approx 0.9772$

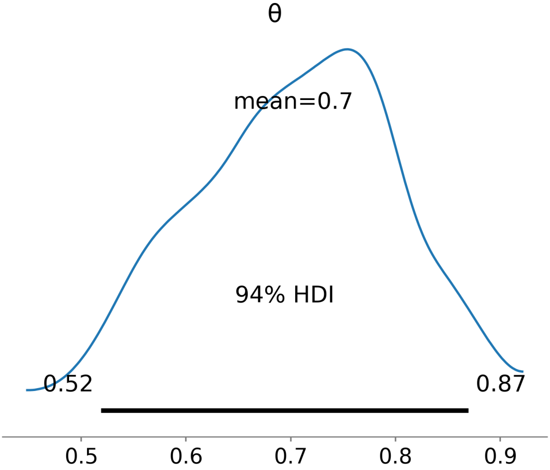
Metropolis-Hasting Algorithm

1. Initialization: Choose an arbitrary point x_t and propose a function $g(x \mid y)$ that satisfies $g(x \mid y) = g(y \mid x)$
2. For each iteration t:
   - Propose a candidate x' for the next sample by picking from the distribution $g(x' \mid x_t)$
   - Calculate the acceptance ratio $\alpha = f(x')/f(x_t) = P(x')/P(x_t)$ (f is proportional to the density of P)
   - Accept or reject:
     - Generate a uniform random number $u \in [0, 1]$
     - If $u \leq \alpha$ then accept the candidate by setting $x_{t+1} = x'$
     - If $u > \alpha$ then reject the candidate and set $x_{t+1} = x_t$ instead

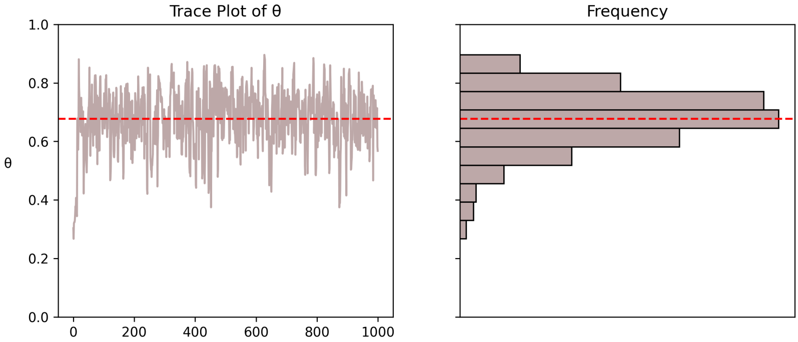
Metropolis-Hasting Algorithm DIY



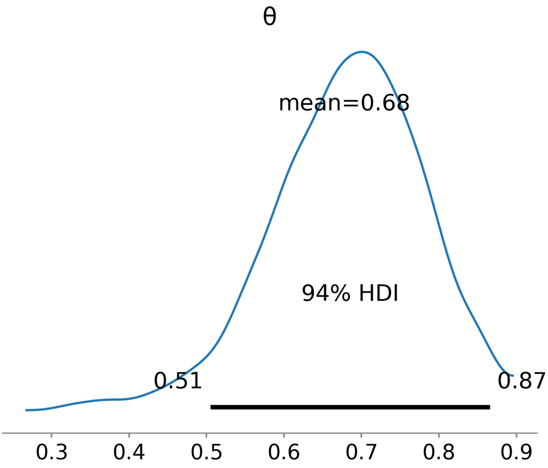
Metropolis-Hasting Algorithm DIY

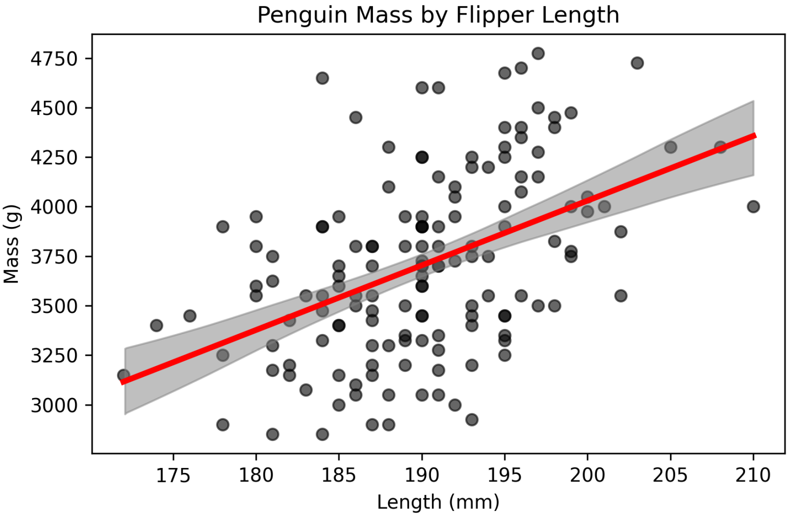


Metropolis-Hasting Algorithm PyMC

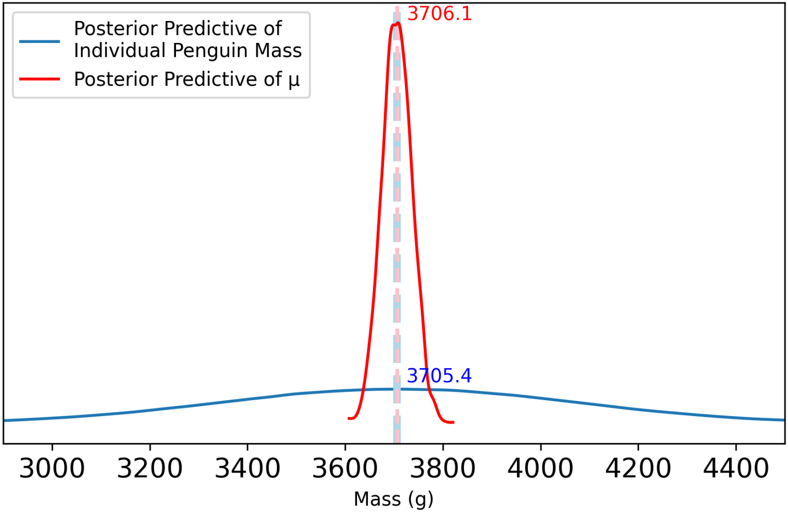


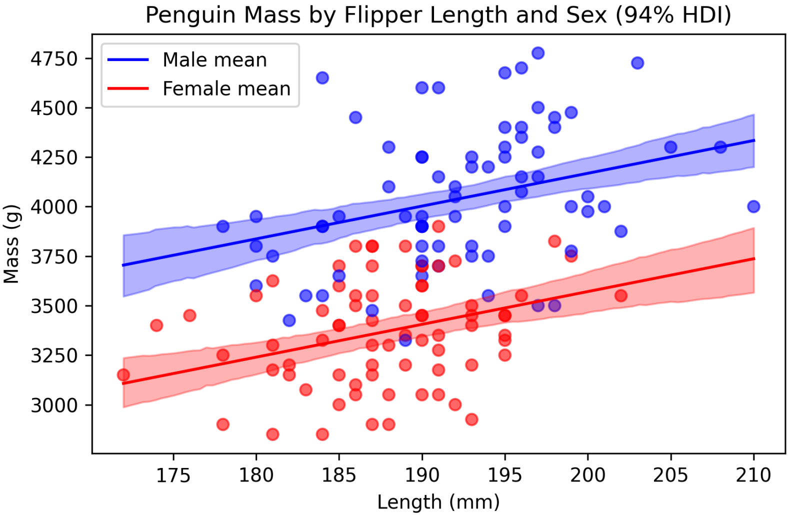
Metropolis-Hasting Algorithm PyMC

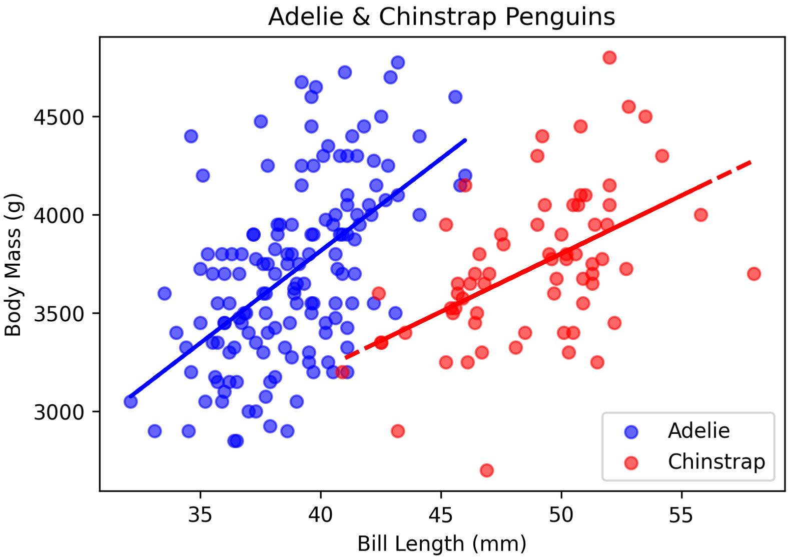



Penguin Mass by Flipper Length
Mass (g)
4750
4500
4250
4000
3750
3500
3250
3000
175
180
185
190
195
200
205
210
Length (mm)

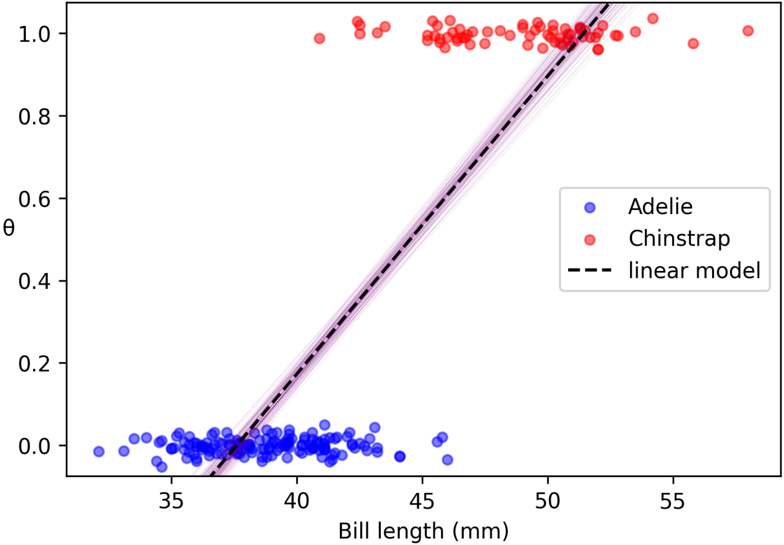
Linear Models



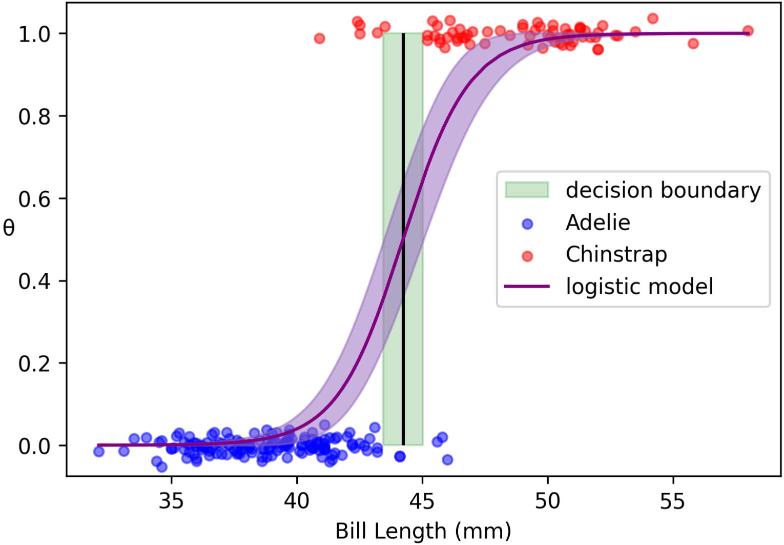

Penguin Mass by Flipper Length and Sex (94% HDI)
Male mean
Female mean
4750
4500
4250
4000
3750
3500
3250
3000
Mass (g)
175
180
185
190
195
200
205
210
Length (mm)


Adelie & Chinstrap Penguins
4500
4000
3500
3000
Body Mass (g)
35
40
45
50
55
Bill Length (mm)
Adelie
Chinstrap

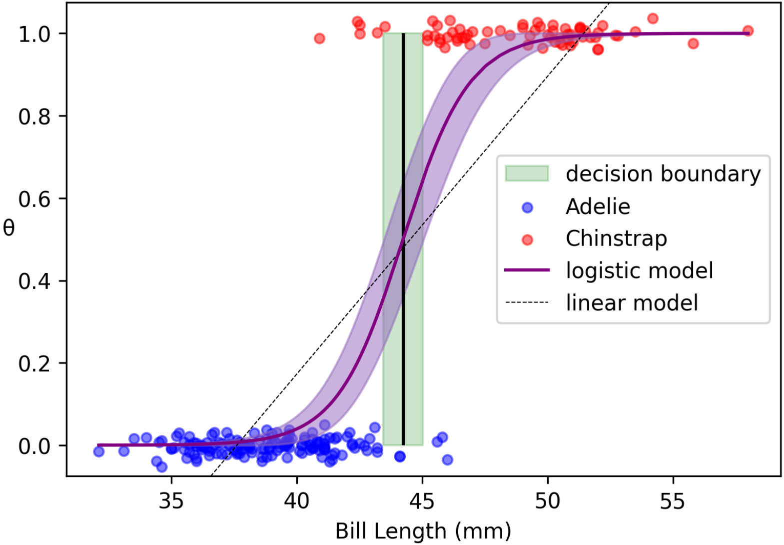
Linear Models



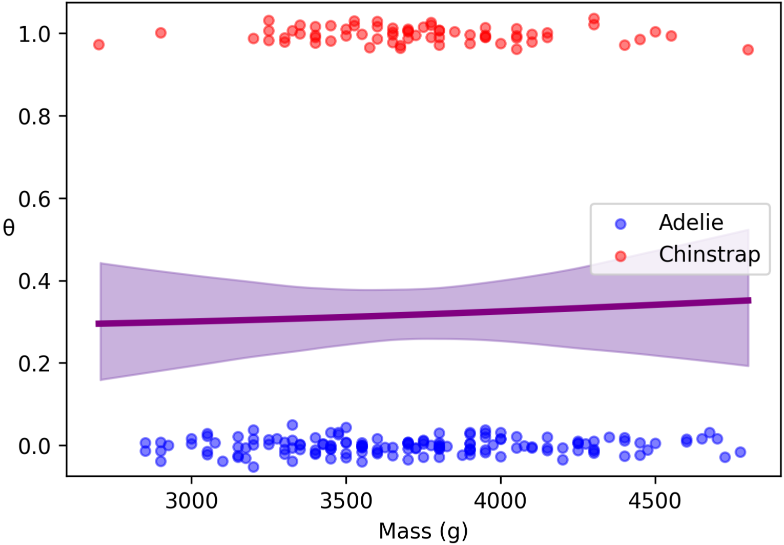
Logistic Models



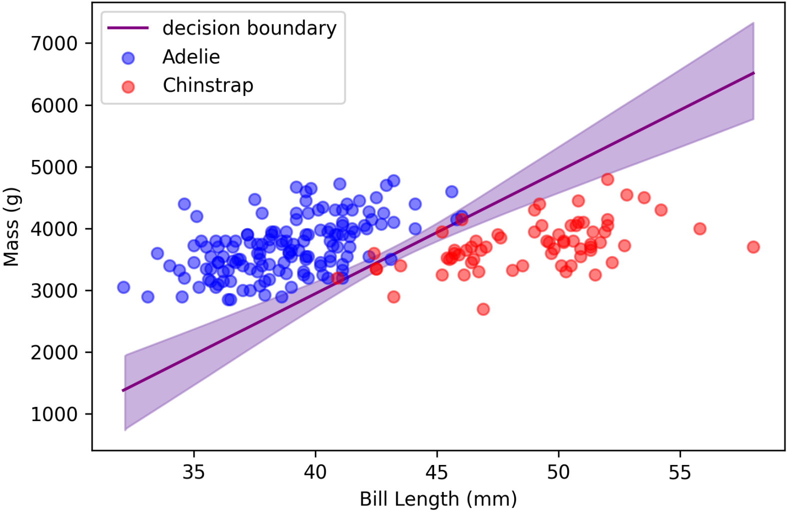
Logistic Models



Logistic Models



Logistic Models



Questions?